

Bilateral Trade with a Lying Cost

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Abstract

Consider the standard bilateral trade setting in Myerson and Satterthwaite 1983, but the buyer and seller have a “lying cost” characterized by a disutility in the form of $\gamma|t - t'|$, where γ is a parameter, t is the agent’s true value, and t' is his/her reported value. I provide some preliminary results on the implementability of the first-best, efficient allocation rule.

1 Introduction

In the bilateral trade setting described in Myerson and Satterthwaite 1983, it is impossible to implement the efficient allocation rule with incentive compatibility, individual rationality, and budget balance. This assumes that agents treat their report as a strategic decision concerning only their possible allocation and payment. I consider a setting where all agents have a commonly-known lying cost, $\gamma|t - t'|$. With a sufficiently high lying cost $\gamma \geq 0.5$, fully efficient trade is implementable for any value distributions. With $\gamma \geq 0.25$, fully efficient trade is implementable for any shared value distributions.

There are many possible explanations for why someone may be averse to lying or inclined to truth-telling. Kartik 2009 presents a similar lying cost in a strategic communication setting. Some motivations for introducing a lying cost in that setting, such as potential ex-post verification or an innate behavioral aversion to lying, can be transferred to this setting.

Saran 2011 considers the bilateral trade setting with the presence of “naive” agents. These agents are committed to truthful reporting, while the other “strategic” agents act strategically. In this setting, the mechanism designer can improve efficiency above the usual Myerson-Satterthwaite level by extracting rents from naive agents to compensate strategic ones. The setting I consider in this paper can be seen somewhat as a generalization, as agents in Saran 2011 have either $\gamma = 0$ (strategic) or $\gamma = \infty$ (naive). However, this paper will not consider γ as a type agents may realize.

It is important to note that, for my results, I restrict attention to direct bargaining mechanisms without invoking a revelation principle. This is because there is no reasonable definition of lying in an arbitrary indirect mechanism, such as an ascending auction, and thus no reasonable way to attribute a cost to lying.

The sections will be as follows. Section 2 will describe the model I use and important definitions, Sections 3, 4, and 5 will discuss incentive compatibility, individual rationality,

and budget balance in that order. Section 6 will be the main results section discussing the efficient allocation. Section 7 will give concluding remarks.

2 The Model

We follow the setup in Myerson and Satterthwaite 1983, but alter the payoff definitions to include the lying cost. The convention throughout this paper will be to denote the buyer's value(unrealized, realized) by V, v and the seller's value(unrealized, realized) by C, c and the buyer's functions with a subscript B and the seller's functions with a subscript S . The buyer and seller together will be referred to generally as "agents." Denote the following:

$$\begin{array}{ll} F_B, \text{ the buyer's value distribution} & F_S, \text{ the seller's value distribution} \\ f_B, \text{ the density of } F_B & f_S, \text{ the density of } F_S \\ [\underline{v}, \bar{v}], \text{ the support of } F_B & [\underline{c}, \bar{c}], \text{ the support of } F_S \end{array}$$

We will identify a distribution by its cumulative distribution function, so F_B and F_S are more specifically the value distributions' CDFs. The buyer and seller's values V and C are drawn independently from these distributions. f_B and f_S are assumed to be continuous and strictly-positive over their supports. Their supports are also assumed to intersect at an interval of length greater than 0. Next, denote the mechanisms rules:

$$\begin{array}{ll} q(v, c), \text{ the allocation rule or probability of trade} \\ p_B(v, c), \text{ the buyer's payment rule} & p_S(v, c), \text{ the seller's payment rule} \end{array}$$

Note that we do not require $p_B = p_S$. Any difference at the end of the trade will go to or be taken from an intermediary. For this paper, we will consider ourselves, the mechanism designer, the intermediary.

For the next quantities and throughout this paper, we will denote a buyer's report as v' and a seller's report as c' . Then, denote the interim values given the other agent reports truthfully:

$$\begin{aligned} Q_B(v') &= \int_{\underline{c}}^{\bar{c}} q(v', c) f_S(c) dc, & Q_S(c') &= \int_{\underline{v}}^{\bar{v}} q(v, c') f_B(v) dv \\ P_B(v') &= \int_{\underline{c}}^{\bar{c}} p_B(v', c) f_S(c) dc, & P_S(c') &= \int_{\underline{v}}^{\bar{v}} p_S(v, c') f_B(v) dv \end{aligned}$$

$$U_B(v, v') = vQ_B(v') - P_B(v') - \gamma|v - v'|, \quad U_S(c, c') = P_S(c') - cQ_S(c') - \gamma|c - c'|$$

where $\gamma \geq 0$. So, Q gives the agent's interim probability of trade given his/her report, P gives the agent's interim expected payment given his/her report, and U gives the agent's interim expected payoff given his/her true value and report. For compactness, we will write $U(v, v)$ as $U(v)$.

With these, we can define important terms to be used throughout this paper.

Definition 1. A *direct bargaining mechanism* (or *mechanism* for short, in this paper) is a triple that consists of an allocation rule q , the buyer's payment rule p_B , and the seller's payment rule p_S .

I reiterate that we will only consider direct bargaining mechanisms without using the revelation principle as justification because “lying” is unreasonable to define over all possible mechanisms.

Definitions 2, 3, 4, and 5 describe desirable qualities of a mechanism.

Definition 2. A mechanism is *incentive compatible (IC)* if and only if reporting truthfully is an equilibrium for agents. That is, for all $v, v' \in [\underline{v}, \bar{v}]$, $U_B(v) \geq U_B(v, v')$ and for all $c, c' \in [\underline{c}, \bar{c}]$, $U_S(c) \geq U_S(c, c')$.

The second sentence states that when the other agent reports truthfully, the first agent finds it optimal to also report truthfully. We will call such an equilibrium a *truthful equilibrium*.

Definition 3. A mechanism is *individually rational (IR)* if and only if agents' expected utility in truthful equilibrium after learning their value is greater than or equal to 0.

For an incentive compatible mechanism, individual rationality is satisfied if and only if for all $v, v' \in [\underline{v}, \bar{v}]$, $U_B(v) \geq 0$ and for all $c, c' \in [\underline{c}, \bar{c}]$, $U_S(c) \geq 0$. IC and IR are an incentive constraint and a participation constraint, respectively. Incentive compatibility requires that agents, in some way, find it optimal to report truthfully to the mechanism. IR requires that agents find it optimal to participate in the first place after learning their value.

Definition 4. A mechanism is *budget balanced (BB)* if and only if its expected revenue in truthful equilibrium is greater than or equal to 0. That is, $\mathbb{E}[p_B(V, C) - p_S(V, C)] \geq 0$.

This quality is desirable because as a designer, we would like to not have to subsidize the trade. Note this definition is specifically for ex-ante budget balance. We do not lose anything by restricting our results to this version of budget balance. The result by Borgers and Norman 2009, stating that the ex-ante and ex-post budget balance problems are equivalent, holds in this setting because payments are “detached” from lying costs.

Definition 5. A mechanism is *efficient* if and only if it maximizes expected welfare across the buyer, seller, and designer in truthful equilibrium.

Since payments are fully accounted for (they cannot be created or destroyed), expected welfare is maximized in an IC mechanism if and only if the object is almost surely allocated to the agent that values it more at the end of the mechanism. The efficient allocation rule we will analyze in this paper is $q^*(v, c) = \mathbb{1}(v \geq c)$.

Definition 6. An allocation rule q is *implementable* if and only if there exist payment rules p_B, p_S such that the mechanism (q, p_B, p_S) is IC, IR, and BB.

To analyze these qualities, since the buyer and seller are somewhat symmetrical in this setting, I will explicitly derive conditions for the buyer, and the seller's conditions will follow a similar reasoning, but this reasoning will not be shown explicitly for space's sake.

3 Incentive Compatibility

For incentive compatibility, we start with the condition in its definition. For all $v' \in [\underline{v}, \bar{v}]$ and for all $c' \in [\underline{c}, \bar{c}]$

$$U_B(v) \geq U_B(v, v') \text{ and } U_S(c) \geq U_S(c, c').$$

This means for all $v_1, v_2 \in [\underline{v}, \bar{v}]$ with $v_1 < v_2$,

$$\begin{aligned} U_B(v_1) &\geq U(v_1, v_2) \\ \iff U_B(v_1) &\geq U_B(v_2) - (v_2 - v_1)Q_B(v_2) - \gamma|v_1 - v_2| \end{aligned}$$

and

$$\begin{aligned} U_B(v_2) &\geq U(v_2, v_1) \\ \iff U_B(v_2) &\geq U_B(v_1) - (v_1 - v_2)Q_B(v_1) - \gamma|v_2 - v_1|. \end{aligned}$$

Together, the two inequalities form the condition

$$\begin{aligned} (v_2 - v_1)Q_B(v_1) - \gamma(v_2 - v_1) &\leq U_B(v_2) - U_B(v_1) \leq (v_2 - v_1)Q_B(v_2) + \gamma(v_2 - v_1) \\ \iff (v_2 - v_1)(Q_B(v_1) - \gamma) &\leq U_B(v_2) - U_B(v_1) \leq (v_2 - v_1)(Q_B(v_2) + \gamma) \quad (*) \\ \iff Q_B(v_1) - \gamma &\leq \frac{U_B(v_2) - U_B(v_1)}{v_2 - v_1} \leq Q_B(v_2) + \gamma \end{aligned}$$

Going through the same calculations for the seller, we get a similar condition, for all values $c_1 < c_2$:

$$-Q_S(c_1) - \gamma \leq \frac{U_S(c_2) - U_S(c_1)}{c_2 - c_1} \leq -Q_S(c_2) + \gamma$$

Thus, by taking limits with $v_1 \rightarrow v_2$, where U_B and U_S are differentiable, their derivatives must satisfy

$$Q_B(v) - \gamma \leq U'_B(v) \leq Q_B(v) + \gamma \text{ and } -Q_S(c) - \gamma \leq U'_S(c) \leq -Q_S(c) + \gamma.$$

These derivative conditions are necessary but not sufficient, generally. When we take limits to pass to the derivative, we lose our chain of equivalences. Notice that (*) implies Lipschitz continuity with Lipschitz constant $1 + \gamma$ for U_B since Q_B is at most 1 and at least 0. Importantly, this means U_B is absolutely continuous so is integrable and satisfies the fundamental theorem of calculus. Additionally, absolute continuity implies differentiability almost everywhere, so its derivative condition is satisfied almost everywhere. A similar reasoning shows this for U_S and its derivative condition.

For a sufficient condition, suppose Q_B is weakly increasing and U_B is absolutely continuous and satisfies its derivative condition almost everywhere. Integrating the derivative condition over $[v_1, v_2]$ with $v_1 < v_2$ gives

$$\begin{aligned} \int_{v_1}^{v_2} Q_B(v) - \gamma dv &\leq \int_{v_1}^{v_2} U'_B(v) dv \leq \int_{v_1}^{v_2} Q_B(v) + \gamma dv \\ \implies (v_2 - v_1)(Q_B(v_1) - \gamma) &\leq U_B(v_2) - U_B(v_1) \leq (v_2 - v_1)(Q_B(v_2) + \gamma) \end{aligned}$$

since Q_B is weakly increasing. This is the same condition as (*). Likewise, if Q_S is weakly decreasing and U_S is absolutely continuous and satisfies its derivative condition almost everywhere, we have the seller's version of this condition by a similar reasoning. Summarizing,

Proposition 1. *If Q_B is weakly increasing and Q_S is weakly decreasing, then a mechanism is incentive compatible if and only if U_B and U_S are absolutely continuous and for almost every $v \in [\underline{v}, \bar{v}]$ and for almost every $c \in [\underline{c}, \bar{c}]$,*

$$Q_B(v) - \gamma \leq U'_B(v) \leq Q_B(v) + \gamma \text{ and } -Q_S(c) - \gamma \leq U'_S(c) \leq -Q_S(c) + \gamma. \quad (1)$$

4 Individual Rationality

Now, suppose we have an incentive compatible mechanism with Q_B weakly increasing and Q_S weakly decreasing. Individual rationality requires for all $v \in [\underline{v}, \bar{v}]$ and for all $c \in [\underline{c}, \bar{c}]$,

$$U_B(v) \geq 0 \text{ and } U_S(c) \geq 0.$$

Since our mechanism is incentive compatible, (1) implies for almost every $v \in [\underline{v}, \bar{v}]$,

$$U'_B(v) \geq Q_B(v) - \gamma.$$

Integrating over $[v_1, v_2]$ with $v_1 < v_2$,

$$\begin{aligned} \int_{v_1}^{v_2} U'_B(v) dv &\geq \int_{v_1}^{v_2} Q_B(v) - \gamma dv \\ \implies U_B(v_2) - U_B(v_1) &\geq \int_{v_1}^{v_2} Q_B(v) - \gamma dv \\ \implies U_B(v_2) &\geq \int_{v_1}^{v_2} Q_B(v) - \gamma dv, \end{aligned}$$

since $U_B(v_1) \geq 0$. Since our selection of $v_1 < v_2$ was arbitrary,

$$U_B(v_2) \geq \sup_{v_1 \in [\underline{v}, v_2]} \int_{v_1}^{v_2} Q_B(v) - \gamma dv. \quad (**)$$

If on $[\underline{v}, v_2]$ $Q_B(v) - \gamma$ is sometimes positive and sometimes negative our optimal choice v_1^* is a point such that before it, $Q_B(v) - \gamma \leq 0$ and after it $Q_B(v) - \gamma \geq 0$ which exists since Q_B is weakly increasing. If $Q_B(v) - \gamma$ is always nonnegative, $v_1^* = \underline{v}$ and if $Q_B(v) - \gamma$ is always negative, $v_1^* = v_2$. Thus,

$$\sup_{v_1 \in [\underline{v}, v_2]} \int_{v_1}^{v_2} Q_B(v) - \gamma dv = \int_{\underline{v}}^{v_2} \max \{Q_B(v) - \gamma, 0\} dv.$$

Again, since our selection of $v_1 < v_2$ was arbitrary, (**) and the above give for all $v \in [\underline{v}, \bar{v}]$,

$$U_B(v) \geq \int_{\underline{v}}^v \max \{Q_B(t) - \gamma, 0\} dt$$

A similar reasoning for the seller gives for all $c \in [\underline{c}, \bar{c}]$

$$U_S(c) \geq \int_c^{\bar{c}} \max \{Q_S(t) - \gamma, 0\} dt$$

A mechanism that attains equality is also incentive compatible and individually rational. For individual rationality, the integrand is nonnegative, and the limits are in increasing order. For incentive compatibility, notice that when equality is attained, these are integrals of bounded functions and therefore absolutely continuous, and for almost every $v \in [\underline{v}, \bar{v}]$ and for almost every $c \in [\underline{c}, \bar{c}]$,

$$U'_B(v) = \max \{Q_B(v) - \gamma, 0\} \text{ and } U'_S(c) = -\max \{Q_S(c) - \gamma, 0\}.$$

These derivatives satisfy (1) by construction. Thus, we have the following proposition.

Proposition 2. *An incentive compatible mechanism with Q_B weakly increasing and Q_S weakly decreasing is individually rational if and only if for all $v \in [\underline{v}, \bar{v}]$ and for all $c \in [\underline{c}, \bar{c}]$*

$$U_B(v) \geq \int_{\underline{v}}^v \max \{Q_B(t) - \gamma, 0\} dt \text{ and } U_S(c) \geq \int_c^{\bar{c}} \max \{Q_S(t) - \gamma, 0\} dt.$$

In particular, a mechanism with Q_B weakly increasing and Q_S weakly decreasing and

$$U_B(v) = \underline{U}_B(v) := \int_{\underline{v}}^v \max \{Q_B(t) - \gamma, 0\} dt, \quad U_S(c) = \underline{U}_S(c) := \int_c^{\bar{c}} \max \{Q_S(t) - \gamma, 0\} dt.$$

is incentive compatible and individually rational.

5 Budget Balance

Given an allocation rule and interim utilities, we can express expected revenue under truthful reporting without explicitly specifying payments first. Recall that, by definition, $P_B(v) = vQ_B(v) - U_B(v)$ and $P_S(c) = cQ_S(c) + U_S(c)$, so

$$\begin{aligned} \mathbb{E}[p_B(V, C)] - \mathbb{E}[p_S(V, C)] &= \mathbb{E}[P_B(V)] - \mathbb{E}[P_S(C)] \\ &= \mathbb{E}[VQ_B(V) - U_B(V)] - \mathbb{E}[CQ_S(C) + U_S(C)] \\ &= \mathbb{E}[VQ_B(V)] - \mathbb{E}[U_B(V)] - \mathbb{E}[CQ_S(C)] - \mathbb{E}[U_S(C)] \\ &= \mathbb{E}[Vq(V, C)] - \mathbb{E}[U_B(V)] - \mathbb{E}[Cq(V, C)] - \mathbb{E}[U_S(C)] \\ &= \mathbb{E}[(V - C)q(V, C)] - (\mathbb{E}[U_B(V)] + \mathbb{E}[U_S(C)]). \end{aligned}$$

In the final expression, the first term is expected gains from trade. The second is the agents' expected utility. With a fixed allocation rule, expected gains from trade is also fixed, but we have some freedom over how much utility we give to the agents. To maximize expected revenue and give us the best chance of budget balance, we should choose the lowest

interim utilities to give to the agents while preserving incentive compatibility and individual rationality. By Proposition 2, for an IC, IR mechanism with Q_B weakly increasing and Q_S weakly decreasing these are $\underline{U}_B$ and $\underline{U}_S$.

Given an allocation rule with Q_B weakly increasing and Q_S weakly decreasing these interim utilities are also achievable by setting $p_B(v, c) = vq(v, c) - \underline{U}_B(v)$ and $p_S(v, c) = cq(v, c) + \underline{U}_S(c)$. These payments set interim payments, and thus expected revenue, as above. Furthermore, by Proposition 2, this mechanism is IC, IR since it achieves the interim utilities $\underline{U}_B$ and $\underline{U}_S$.

Then, we have that an allocation rule with Q_B weakly increasing and Q_S weakly decreasing is implementable if and only if

$$\mathbb{E}[(V - C)q(V, C)] - (\mathbb{E}[\underline{U}_B(V)] + \mathbb{E}[\underline{U}_S(C)]) \geq 0.$$

For the buyer,

$$\begin{aligned} \mathbb{E}[\underline{U}_B(V)] &= \int_{\underline{v}}^{\bar{v}} \left(\int_{\underline{v}}^v \max\{Q_B(t) - \gamma, 0\} dt \right) f_B(v) dv \\ &= \int_{\underline{v}}^{\bar{v}} \int_{\underline{v}}^v \max\{Q_B(t) - \gamma, 0\} f_B(v) dt dv \\ &= \int_{\underline{v}}^{\bar{v}} \int_t^{\bar{v}} \max\{Q_B(t) - \gamma, 0\} f_B(v) dv dt, \text{ by Fubini's theorem} \\ &= \int_{\underline{v}}^{\bar{v}} \max\{Q_B(t) - \gamma, 0\} \int_t^{\bar{v}} f_B(v) dv dt \\ &= \int_{\underline{v}}^{\bar{v}} \max\{Q_B(t) - \gamma, 0\} (1 - F_B(t)) dt \\ &= \int_{\underline{v}}^{\bar{v}} \max\{Q_B(v) - \gamma, 0\} (1 - F_B(v)) dv \end{aligned}$$

For the use of Fubini's theorem, notice f_B is assumed to be continuous on a compact interval so is bounded, and $\max\{Q_B(t) - \gamma, 0\}$ is also bounded. Then, the integrand is absolutely integrable so switching integration order is justified.

A similar calculation for the seller gives

$$\mathbb{E}[\underline{U}_S(C)] = \int_{\underline{c}}^{\bar{c}} \max\{Q_S(c) - \gamma, 0\} F_S(c) dc.$$

And thus we have,

Proposition 3. *An allocation rule with Q_B weakly increasing and Q_S weakly decreasing is implementable if and only if*

$$\mathbb{E}[(V - C)q(V, C)] - (\mathbb{E}[\underline{U}_B(V)] + \mathbb{E}[\underline{U}_S(C)]) \geq 0.$$

Furthermore,

$$\begin{aligned}\mathbb{E}[\underline{U}_B(V)] &= \int_{\underline{v}}^{\bar{v}} \max\{Q_B(v) - \gamma, 0\}(1 - F_B(v)) dv, \\ \mathbb{E}[\underline{U}_S(C)] &= \int_{\underline{c}}^{\bar{c}} \max\{Q_S(c) - \gamma, 0\}F_S(c) dc.\end{aligned}$$

6 The Efficient Allocation

The efficient allocation rule $q^*(v, c) = \mathbb{1}(v \geq c)$ has interim allocation probabilities $Q_B(v) = F_S(v)$ and $Q_S(c) = 1 - F_B(c)$. We can calculate the efficient allocation's expected gains from trade G^* as a function of F_B and F_S .

$$\begin{aligned}G^*(F_B, F_S) &= \mathbb{E}[(V - C)\mathbb{1}(V \geq C)] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (v - c)\mathbb{1}(v \geq c)f_S(c)f_B(v) dc dv \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \mathbb{1}(c \leq t)\mathbb{1}(t \leq v) dt \right) f_S(c)f_B(v) dc dv \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \mathbb{1}(c \leq t)f_S(c) dc \right) \left(\int_{-\infty}^{\infty} \mathbb{1}(t \leq v)f_B(v) dv \right) dt \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^t f_S(c) dc \right) \left(\int_t^{\infty} f_B(v) dv \right) dt \\ &= \int_{-\infty}^{\infty} F_S(t)(1 - F_B(t)) dt.\end{aligned}$$

For the third equality, recall that the layer-cake representation of $v - c$ when $v \geq c$ is

$$v - c = \int_c^v dt = \int_{-\infty}^{\infty} \mathbb{1}(c \leq t \leq v) dt = \int_{-\infty}^{\infty} \mathbb{1}(c \leq t)\mathbb{1}(t \leq v) dt.$$

And the fourth equality is justified by Fubini's theorem again.

By Proposition 3, the agents' minimum expected utility (while remaining IC, IR) U^* as a function of γ , F_B , and F_S is

$$U^*(\gamma, F_B, F_S) = \int_{\underline{v}}^{\bar{v}} \max\{F_S(v) - \gamma, 0\}(1 - F_B(v)) dv + \int_{\underline{c}}^{\bar{c}} \max\{1 - F_B(c) - \gamma, 0\}F_S(c) dc.$$

Since $F_S(v)$ is weakly increasing and $1 - F_B(c)$ weakly decreasing by definition, we can use Proposition 3. Then, the efficient allocation rule is implementable if and only if

$$G^*(F_B, F_S) - U^*(\gamma, F_B, F_S) \geq 0 \tag{3}$$

U^* is weakly decreasing with γ , so the LHS of (3) is weakly increasing with γ . Thus, if (3) is satisfied for γ_1 , it is also satisfied for all $\gamma_2 \geq \gamma_1$.

This first result's reverse direction restates the Myerson-Satterthwaite Theorem.

Proposition 4 (The Myerson-Satterthwaite Theorem). *The efficient allocation rule is not implementable for any distributions if and only if $\gamma = 0$.*

Proof. Start with the reverse direction. For some distributions F_B and F_S , agents' minimum expected utility when $\gamma = 0$ is

$$\begin{aligned} U^*(0, F_B, F_S) &= \int_{\underline{v}}^{\bar{v}} \max\{F_S(v) - 0, 0\}(1 - F_B(v)) dv + \int_{\underline{c}}^{\bar{c}} \max\{1 - F_B(c) - 0, 0\}F_S(c) dc \\ &= \int_{\underline{v}}^{\bar{v}} F_S(v)(1 - F_B(v)) dv + \int_{\underline{c}}^{\bar{c}} (1 - F_B(c))F_S(c) dc, \end{aligned}$$

and expected gains from trade are

$$G^*(F_B, F_S) = \int_{-\infty}^{\infty} F_S(t)(1 - F_B(t)) dt.$$

So (3) is,

$$\int_{-\infty}^{\infty} F_S(t)(1 - F_B(t)) dt - \int_{\underline{v}}^{\bar{v}} F_S(v)(1 - F_B(v)) dv - \int_{\underline{c}}^{\bar{c}} (1 - F_B(c))F_S(c) dc \geq 0.$$

Notice the integrands for these three terms are the same. However, the integral on the intersection of $[\underline{v}, \bar{v}]$ and $[\underline{c}, \bar{c}]$, which is strictly positive by our assumptions, is subtracted out twice. Thus, the LHS must be strictly negative, so the efficient allocation is not implementable.

For the forward direction, let $\gamma > 0$ be fixed. Our goal is to find distributions F_B, F_S such that we have (3). Consider when the buyer's value distribution is Uniform $[0, 1]$ and the seller's is Uniform $[1 - \epsilon, 2 - \epsilon]$ for some $0 < \epsilon \leq 1$. The intersection of the two supports is the interval $[1 - \epsilon, 1]$, so these are valid distributions.

To compute gains from trade, note the CDFs of these uniform distributions are

$$F_B(t) = \begin{cases} 0 & \text{if } t < 0 \\ t & \text{if } 0 \leq t \leq 1 \\ 1 & \text{if } t > 1 \end{cases} \text{ and } F_S(t) = \begin{cases} 0 & \text{if } t < 1 - \epsilon \\ t - (1 - \epsilon) & \text{if } 1 - \epsilon \leq t \leq 2 - \epsilon \\ 1 & \text{if } t > 2 - \epsilon \end{cases}.$$

$$\begin{aligned} G_\epsilon^*(F_B, F_S) &= \int_{1-\epsilon}^1 (t - (1 - \epsilon))(1 - t) dt \\ &= \int_0^\epsilon x(\epsilon - x) dx, \quad x = t - (1 - \epsilon) \\ &= \frac{\epsilon x^2}{2} - \frac{x^3}{3} \Big|_{x=0}^\epsilon \\ &= \frac{\epsilon^3}{6}. \end{aligned}$$

Agents' minimum expected utility when $\gamma \leq \epsilon$ is

$$\begin{aligned}
& U^*(\gamma, F_B, F_S) \\
&= \int_0^1 \max\{F_S(v) - \gamma, 0\}(1 - F_B(v)) dv + \int_{1-\epsilon}^{2-\epsilon} \max\{1 - F_B(c) - \gamma, 0\}F_S(c) dc \\
&= \int_{1-\epsilon}^1 \max\{(v - (1 - \epsilon) - \gamma)(1 - v), 0\} dv + \int_{1-\epsilon}^1 \max\{(1 - c - \gamma)(c - (1 - \epsilon)), 0\} dc \\
&= 2 \int_{1-\epsilon+\gamma}^1 (1 - t)(t - (1 - \epsilon) - \gamma) dt, \quad t = 2 - \epsilon - c \\
&= 2 \int_{\gamma}^{\epsilon} (\epsilon - x)(x - \gamma) dx, \quad x = t - (1 - \epsilon) \\
&= 2 \left(\frac{\epsilon x^2}{2} - \frac{x^3}{3} - \gamma \epsilon x + \frac{\gamma x^2}{2} \right) \Big|_{x=\gamma}^{\epsilon} \\
&= \frac{(\epsilon - \gamma)^3}{3}.
\end{aligned}$$

If $\gamma > \epsilon$, $U^*(\gamma, F_B, F_S) = 0$. Then, choose ϵ so that $\frac{(\epsilon - \gamma)^3}{3} \leq \frac{\epsilon^3}{6} \iff \gamma \geq \epsilon \left(1 - \sqrt[3]{\frac{1}{2}}\right)$, making $G^*(F_B, F_S) \geq U^*(\gamma, F_B, F_S)$, satisfying (3). $\square$

From this proof, we can establish that the minimum γ such that the efficient allocation is implementable for Uniform $[0, 1]$ distributions is $\gamma^* = \left(1 - \sqrt[3]{\frac{1}{2}}\right)$.

Next, we establish another threshold for γ .

Proposition 5. *For $\gamma \geq \frac{1}{2}$, for all F_B, F_S , the mechanism $(\mathbb{1}(v \geq c), \frac{v+c}{2}\mathbb{1}(v \geq c), \frac{v+c}{2}\mathbb{1}(v \geq c))$ is incentive compatible, individually rational, budget balanced, and efficient.*

Proof. Individual rationality, budget balance, and efficiency are clear (even in their strongest senses). It remains to check incentive compatibility.

Fix the seller's report c' , and consider the buyer's decision, denoting his payoff based on his value and report and the seller's report $u_B(v, v', c')$. When the buyer has value v , the difference in payoff between reporting v and reporting v' is

$$\begin{aligned}
& u_B(v, v, c') - u_B(v, v', c') \\
&= \mathbb{1}(v \geq c') \left(v - \frac{v + c'}{2} \right) - \left(\mathbb{1}(v' \geq c') \left(v - \frac{v' + c'}{2} \right) - \gamma |v - v'| \right) \\
&= \mathbb{1}(v \geq c') \frac{v - c'}{2} - \left(\mathbb{1}(v' \geq c') \left(\frac{v - v'}{2} + \frac{v - c'}{2} \right) - \gamma |v - v'| \right) \\
&= \frac{v - c'}{2} (\mathbb{1}(v \geq c') - \mathbb{1}(v' \geq c')) - \left(\mathbb{1}(v' \geq c') \left(\frac{v - v'}{2} \right) - \gamma |v - v'| \right).
\end{aligned}$$

The second term is weakly negative because $\gamma \geq \frac{1}{2}$, so subtracting it contributes weakly positively. For the first term, suppose $v < v'$. Consider three cases:

- $c' \leq v < v'$: the first term equals $\frac{v-c'}{2}(1-1) = 0$.
- $v < c' \leq v'$: the first term equals $0 - \frac{v-c'}{2} > 0$.
- $v < v' < c'$: the first term equals $\frac{v-c'}{2}(0-0) = 0$.

In any case, the first term is greater than or equal to 0. A similar reasoning shows this for when $v' < v$. Thus, it is a weakly dominant strategy to report truthfully, so the mechanism is incentive compatible. $\square$

So, when we have a sufficiently high lying cost, we can safely use the natural “trade if efficient and split the surplus” mechanism analyzed in Chatterjee and Samuelson 1983. In fact, $\gamma = \frac{1}{2}$ is the greatest lower-bound for the efficient allocation to be implementable across all distributions.

Corollary 1. *The efficient allocation rule is implementable for all distributions if and only if $\gamma \geq \frac{1}{2}$.*

Proof. The reverse direction follows directly from Proposition 5.

For the forward direction, let $\gamma < \frac{1}{2}$ be fixed. Consider sequences of valid densities on $[0, 1]$ that become increasingly concentrated at 1 for the buyer and at 0 for the seller. An example of such is $f_{B,k}(v) = h_k(1-v)$ and $f_{S,k}(c) = h_k(c)$ where

$$h_k(t) = \frac{ke^{-kt}}{1 - e^{-k}}.$$

For all $t \in (0, 1)$, as $k \rightarrow \infty$, $F_{S,k}(t) \rightarrow 1$ and $F_{B,k}(t) \rightarrow 0$. Notice that the integrands of G_k^* and U_k^* are uniformly bounded by 1 and 2, respectively, for all k and these bounds are clearly integrable. Then, we are justified in using the Dominated Convergence Theorem to exchange limits with integration. We get that gains from trade $G_k^* \rightarrow 1$ and agents' minimum expected utility $U_k^* \rightarrow 2(1-\gamma)$. Since $\gamma < \frac{1}{2}$, $1 < 2(1-\gamma)$. Then, choose k large enough that $G_k^* < U_k^*$. $\square$

Proposition 6. *The efficient allocation rule is implementable for all distributions with $F_B = F_S$ if and only if $\gamma \geq \frac{1}{4}$.*

Proof. Suppose $F_B = F_S = F$ with support $[\underline{t}, \bar{t}]$. Start with the reverse direction. It suffices to show that the efficient allocation rule is implementable for $\gamma = 1/4$. Gains from trade and agents' minimum expected utility is

$$G^*(F, F) = \int_{\underline{t}}^{\bar{t}} F(t)(1 - F(t)) dt$$

$$U^*\left(\frac{1}{4}, F, F\right) = \int_{\underline{t}}^{\bar{t}} \max\left\{F(t) - \frac{1}{4}, 0\right\}(1 - F(t)) dt + \int_{\underline{t}}^{\bar{t}} \max\left\{1 - F(t) - \frac{1}{4}, 0\right\}F(t) dt.$$

Since the limits of all these integrals are the same, compare the integrands of G^* and U^* in two cases:

- $F(t) \in [\frac{1}{4}, \frac{3}{4}]$: both of the left expressions in $\max\{\}$ are always weakly positive.

$$\text{Integrand of } G^* = F(t) - F(t)^2$$

$$\begin{aligned} \text{Integrands of } U^* &= (F(t) - \frac{1}{4})(1 - F(t)) + (1 - F(t) - \frac{1}{4})F(t) \\ &= \left(F(t) - \frac{1}{4} - F(t)^2 + \frac{1}{4}F(t)\right) + \left(F(t) - F(t)^2 - \frac{1}{4}F(t)\right) \\ &= 2F(t) - 2F(t)^2 - \frac{1}{4} \\ \text{Difference} &= -F(t) + F(t)^2 + \frac{1}{4} \\ &= \left(F(t) - \frac{1}{2}\right)^2 \geq 0 \end{aligned}$$

- $F(t) \notin [\frac{1}{4}, \frac{3}{4}]$: one of the left expressions in $\max\{\}$ is always weakly negative.

$$\text{Integrand of } G^* = F(t) - F(t)^2$$

$$\text{Integrand of } U^* = \begin{cases} (1 - F(t) - \frac{1}{4})F(t) & \text{if } F(t) < \frac{1}{4} \\ (F(t) - \frac{1}{4})(1 - F(t)) & \text{if } F(t) > \frac{3}{4} \end{cases}$$

In any case, the integrand of U^* is less than or equal to the integrand of G^* .

For the forward direction, we will use a similar strategy as in the proof of Corollary 1. Fix $\gamma < \frac{1}{4}$ and let h_k be defined as it is in that proof. Consider $f_{B,k}(t) = f_{S,k}(t) = f_k(t) = \frac{h_k(t) + h_k(1-t)}{2}$. Here, densities become increasingly concentrated at the endpoints for both the buyer and seller. So, for all $t \in (0, 1)$, as $k \rightarrow \infty$, $F_k(t) \rightarrow \frac{1}{2}$. Using a similar dominated convergence argument, we get that $G_k^* \rightarrow \frac{1}{4}$ and $U_k^* \rightarrow \frac{1}{2} - \gamma$. Since, $\gamma < \frac{1}{4}$, we can choose k large enough that $G_k^* < U_k^*$. $\square$

7 Concluding Remarks

With a sufficiently high, commonly-known cost of lying, the efficient allocation rule can be implemented. The key freedom that is granted from this lying cost is that rents from the first γ of allocation probability allocated can be fully extracted.

Within this setting, we may want to compute second-best allocations, given the first-best efficient allocation is not implementable. A natural extension of this setting would be to include γ as a type that would also need to be screened. It remains to determine how this would be handled. Are more lying-averse types also averse to lying about their lying aversion? Saran 2011 handles this in their two-type setting by forcing naive types to also report that they are naive naively. Strategic types are allowed to report that they are naive, so the mechanism must incentivize them to report that they are strategic instead of naive.

OpenAI’s GPT-6 Astra was very helpful in working out the mathematical details, especially the forward direction proofs of Propositions 4 and 6 and Corollary 1. The overarching mathematical idea used to derive the IC, IR, and BB conditions (using derivatives to determine the shape of minimum interim utilities and expressing payments in terms of those utilities) is fairly derivative from Myerson and Satterthwaite 1983. The document was ”hand”-written (typed into OverLeaf) by me with no AI-generated text.

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